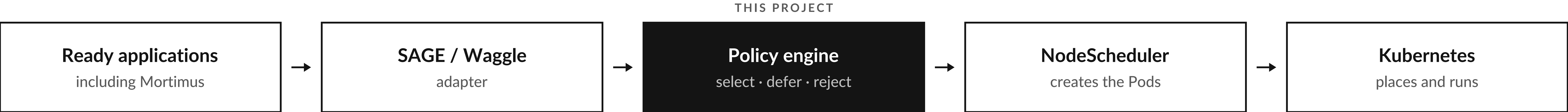


# Resilient urgent orchestration for SAGE

An explainable scheduling policy for SAGE / Waggle nodes

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In urgent computing, the right result delivered **too late** is the wrong result.



SAGE remains the orchestrator — only the selection policy changes. Beehive, WES, the NodeScheduler and Kubernetes are untouched.

| The problem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | How it decides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | What it returns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>SAGE nodes run several applications on limited edge hardware. Routine work and time-sensitive events compete for the same CPU, memory, GPU and execution slots, while priorities change.</p> <p>Three questions must be answered:</p> <div><div>01</div>Which ready application should start next?</div> <div><div>02</div>Can it start without exceeding safe limits?</div> <div><div>03</div>Why was each application selected, deferred or rejected?</div> <hr/> <p>Mortimus — cameras watching a landscape for smoke — is the integration target, not a dependency.</p> | <div><div>01</div>Observe and validate ready and active tasks.</div> <div><div>02</div>Rank valid tasks by priority, deadline slack, queue age and success probability.</div> <div><div>03</div>Admit within concurrency limits, and resource limits when capacity is trustworthy.</div> <div><div>04</div>Explain every selection, deferral or rejection with a stable reason.</div> <div><div>05</div>Degrade to bounded queue order if it fails.</div> <div><div>URGENCY IS DEADLINE SLACK</div><div><b>slack = deadline - now - estimated runtime</b></div></div> | <p>Every candidate leaves with one outcome and a stable reason. One node, one free slot, two tasks:</p> <div><div><div>smoke-detector</div><div>SELECTED</div><div>priority 90 · slack -5 s · score 80.75</div><div>reason: selected · predictedDeadlineMiss: true</div></div><div><div><div>image-sampler</div><div>DEFERRED</div><div>priority 20 · slack +540 s · score 19.65</div><div>reason: concurrency_limit</div></div></div><p>Deferred stays queued for the next free slot; rejected is invalid work.</p></div> |